

Track 2B Curve-First Reranking Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report reranks the already accepted `Track 2` candidate matrix by full-curve validation behavior. It does not execute training, does not alter the dataset structure, and does not provide future curve samples to any model.

- Run Instance: `2026-05-28-19-27-46__track2b_curve_first_reranking`
- Source Track 2 Run: `output\validation_checks\track2_reference_comparison\2026-05-28-12-22-56__track2_full_directional_family_matrix_wave2c_residual_harmonic_temporal_hybrid_track2_refresh_2026_05_28`
- Source Curve Count: `194`
- Source Candidate Count: `111`
- Generated Artifact Directory: `output\validation_checks\track2_curve_first_reranking\2026-05-28-19-27-46__track2b_curve_first_reranking`

Method

The primary ordering key is mean `Track 2` mean-percentage-error over each candidate's valid direction surface. Ties are resolved by P95 mean-percentage-error, worst mean-percentage-error, and mean curve `MAE`. This keeps scalar pointwise registry metrics separate from curve-following evidence.

Available diagnostics from the existing `Track 2` matrix:

- mean curve `MAE` and `RMSE` per operating condition;
- mean percentage error per operating condition;
- P95, worst-condition, and standard-deviation aggregates across conditions.

Deferred diagnostics requiring a future curve-payload export:

- harmonic amplitude and phase error by order;
- derivative or slope continuity error;
- per-revolution residual drift and continuity checks across stitched curves.

Causal Input Boundary

The validation surface is full-curve because the compensation target is continuous `TE` over many consecutive motor revolutions. The runtime input contract remains causal: current point-level operating state, optional short history of already observed samples, or derived causal features only.

Overall Curve-First Leaders

Rank	Candidate	Family	Surface	Direction	Curves	Mean MPE [%]	P95 MPE [%]	Worst MPE [%]
1	rcim_retuned_GBM19_Fw	GBM	Fw	all_valid	97	2.371752	4.911649	5.680916
2	rcim_original_ERT19_Fw	ERT	Fw	all_valid	97	3.252972	6.145160	7.841177
3	rcim_retuned_RF19_Fw	RF	Fw	all_valid	97	3.291549	5.997536	7.048587
4	rcim_original_RF19_Fw	RF	Fw	all_valid	97	3.939860	6.872174	9.475785
5	rcim_original_LGBM19_Fw	LGBM	Fw	all_valid	97	4.016685	10.054446	11.877265
6	rcim_retuned_ERT19_Fw	ERT	Fw	all_valid	97	4.038649	7.599022	9.427596
7	paper_retuned_best_Fw	best_composite	Fw	all_valid	97	4.108527	9.865680	11.737393
8	rcim_retuned_HGBM19_Fw	HGBM	Fw	all_valid	97	4.126307	9.401102	13.400433
9	rcim_retuned_LGBM19_Fw	LGBM	Fw	all_valid	97	4.135160	9.865585	11.737457
10	rcim_retuned_DT19_Fw	DT	Fw	all_valid	97	4.305731	9.062969	12.531740
11	rcim_original_DT19_Fw	DT	Fw	all_valid	97	4.305735	9.062969	12.531740
12	rcim_original_GBM19_Fw	GBM	Fw	all_valid	97	4.312144	8.193384	10.648506
13	rcim_retuned_ET19_Fw	ET	Fw	all_valid	97	4.425530	9.533169	12.508938
14	rcim_original_HGBM19_Fw	HGBM	Fw	all_valid	97	4.493305	10.616761	14.461187
15	rcim_retuned_XGBM19_Fw	XGBM	Fw	all_valid	97	4.587646	10.487803	12.008064

Forward Curve-First Leaders

Rank	Candidate	Family	Surface	Direction	Curves	Mean MPE [%]	P95 MPE [%]	Worst MPE [%]
1	rcim_retuned_GBM19_Fw	GBM	Fw	forward	97	2.371752	4.911649	5.680916
2	rcim_original_ERT19_Fw	ERT	Fw	forward	97	3.252972	6.145160	7.841177
3	rcim_retuned_RF19_Fw	RF	Fw	forward	97	3.291549	5.997536	7.048587
4	rcim_original_RF19_Fw	RF	Fw	forward	97	3.939860	6.872174	9.475785
5	rcim_original_LGBM19_Fw	LGBM	Fw	forward	97	4.016685	10.054446	11.877265
6	rcim_retuned_ERT19_Fw	ERT	Fw	forward	97	4.038649	7.599022	9.427596
7	paper_retuned_best_Fw	best_composite	Fw	forward	97	4.108527	9.865680	11.737393
8	rcim_retuned_HGBM19_Fw	HGBM	Fw	forward	97	4.126307	9.401102	13.400433
9	rcim_retuned_LGBM19_Fw	LGBM	Fw	forward	97	4.135160	9.865585	11.737457
10	rcim_retuned_DT19_Fw	DT	Fw	forward	97	4.305731	9.062969	12.531740
11	rcim_original_DT19_Fw	DT	Fw	forward	97	4.305735	9.062969	12.531740
12	rcim_original_GBM19_Fw	GBM	Fw	forward	97	4.312144	8.193384	10.648506

Backward Curve-First Leaders

Rank	Candidate	Family	Surface	Direction	Curves	Mean MPE [%]	P95 MPE [%]	Worst MPE [%]
1	rcim_retuned_GBM19_Bw	GBM	Bw	backward	97	5.398275	12.280348	29.298779
2	periodic_gru_sequence_Bw	periodic_gru_sequence	Bw	backward	97	5.465946	14.820350	18.759599
3	periodic_gru_sequence_global	periodic_gru_sequence	global	backward	97	6.010388	12.692963	17.220297
4	periodic_lstm_sequence_Bw	periodic_lstm_sequence	Bw	backward	97	6.013045	15.382036	17.697190
5	periodic_lstm_sequence_global	periodic_lstm_sequence	global	backward	97	6.097733	14.674261	17.806152
6	rcim_retuned_ET19_Bw	ET	Bw	backward	97	7.021055	15.287358	28.924505
7	tree_Bw	tree	Bw	backward	97	7.051484	14.115584	15.808625
8	tree_global	tree	global	backward	97	7.118450	13.703254	15.694393
9	rcim_retuned_ERT19_Bw	ERT	Bw	backward	97	7.269106	13.186813	29.804962
10	residual_harmonic_lstm_sequence_sparse_rcim_Bw	residual_harmonic_lstm_sequence_sparse_rcim	Bw	backward	97	7.510266	13.159550	15.012015
11	rcim_retuned_RF19_Bw	RF	Bw	backward	97	7.542522	15.083100	28.325982
12	paper_retuned_best_Bw	best_composite	Bw	backward	97	7.571630	15.644903	29.854827

Surface Leaders

Surface	Leader	Family	Source	Curves	Mean MPE [%]	P95 MPE [%]	Mean Curve MAE [deg]
Bw	rcim_retuned_GBM19_Bw	GBM	rcim_retuned	97	5.398275	12.280348	0.002766
Fw	rcim_retuned_GBM19_Fw	GBM	rcim_retuned	97	2.371752	4.911649	0.001089
global	periodic_lstm_sequence_global	periodic_lstm_sequence	wave2_temporal_entry_registry	194	6.119950	14.716986	0.002707

Scalar Registry Context

- Current scalar registry winner: te_periodic_gru_sequence_remote_Bw from family periodic_gru_sequence_bw .
- Scalar test MAE : 0.002344 and scalar test RMSE : 0.002747 .

The strongest forward curve-first candidate in this reranking is rcim_retuned_GBM19_Fw from family GBM with mean MPE 2.371752 percent and P95 MPE 4.911649 percent. This does not replace the backward or global branches.

Decision

This pass standardizes the curve-first evidence surface and should be read as three parallel selection tracks: `Fw` , `Bw` , and `global` . It does not promote one single program-best model by itself because the real application needs one best candidate per surface and richer harmonic/phase diagnostics still require curve-payload export.

Machine-readable artifacts

- `output\validation_checks\track2_curve_first_reranking\2026-05-28-19-27-46__track2b_curve_first_reranking\candidate_curve_first_ranking.csv`
- `output\validation_checks\track2_curve_first_reranking\2026-05-28-19-27-46__track2b_curve_first_reranking\direction_curve_first_ranking.csv`
- `output\validation_checks\track2_curve_first_reranking\2026-05-28-19-27-46__track2b_curve_first_reranking\track2_curve_first_reranking_summary.yaml`